深度二条件下充分负扭的第一上同调消失

Editorial reading note. Source symbols, including the printed P^1 in the hyperplane paragraph and exponents in the diagram, are preserved; editorial corrections are not inserted into author text.
Counting scope. The primary problem is Tag 0FD8, marked below. All preceding numbered statements are supporting author context and are not counted as additional problems. This is a local pilot with a temporary ID, not production admission.
Copyright (C) 2005–2025 Johan de Jong. Stacks Project authors. Source: Tag 0FD8.
Difficulty H2: 本题在闭点处 depth 至少二时证明充分负的 ample 扭曲具有零 H1。证明先用超平面截面使 H1 维数稳定，再在高次截面的核空间中选避开伴随点的截面，令同一乘法同时为零且单射以迫使稳定维数为零。该反证构造和有限域下降均未被背景直接给出，判 H2。.
History: excerpt from revision local cached snapshot; exact chapter hashes; revision unknown, varieties.tex, lines 11157–11290. Only reference presentation was adapted. Licensed under GNU FDL 1.2 or later; no invariant sections or cover texts. See accompanying COPYING.

Lemma 0.1. *Let k be an infinite field. Let $n \geq 1$. Let $\mathcal{F}$ be a coherent module on $\mathbf{P}^n_k$. Then there exist a nonzero section $s \in \Gamma(\mathbf{P}^n_k, \mathcal{O}(1))$ and a short exact sequence*

$$0 \to \mathcal{F}(-1) \to \mathcal{F} \to i_*\mathcal{G} \to 0$$

where $i : H \to \mathbf{P}^n_k$ is the hyperplane H associated to s and $\mathcal{G} = i^*\mathcal{F}$.

证明. The map $\mathcal{F}(-1) \to \mathcal{F}$ comes from Constructions, Equation (equation multiply on sheaf (Tag 03GJ)) with $n = 1$, $m = -1$ and the section s of $\mathcal{O}(1)$. Let's work out what this map looks like if we restrict it to $D_+(T_0)$. Write $D_+(T_0) = \mathrm{Spec}(k[x_1, \ldots, x_n])$ with $x_i = T_i/T_0$. Identify $\mathcal{O}(1)|_{D_+(T_0)}$ with $\mathcal{O}$ using the section T_0. Hence if $s = \sum a_i T_i$ then $s|_{D_+(T_0)} = a_0 + \sum a_i x_i$ with the identification chosen above. Furthermore, suppose $\mathcal{F}|_{D_+(T_0)}$ corresponds to the finite $k[x_1, \ldots, x_n]$-module M. Via the identification $\mathcal{F}(-1) = \mathcal{F} \otimes \mathcal{O}(-1)$ and our chosen trivialization of $\mathcal{O}(1)$ we see that $\mathcal{F}(-1)$ corresponds to M as well. Thus restricting $\mathcal{F}(-1) \to \mathcal{F}$ to $D_+(T_0)$ gives the map

$$M \xrightarrow{a_0 + \sum a_i x_i} M$$

To see that the arrow is injective, it suffices to pick $a_0 + \sum a_i x_i$ outside any of the associated primes of M, see Algebra, Lemma lemma ass zero divisors (Tag 00LD). By Algebra, Lemma lemma finite ass (Tag 00LC) the set $\mathrm{Ass}(M)$ of associated primes of M is finite. Note that for $\mathfrak{p} \in \mathrm{Ass}(M)$ the intersection $\mathfrak{p} \cap \{a_0 + \sum a_i x_i\}$ is a proper k-subvector space. We conclude that there is a finite family of proper sub vector spaces $V_1, \ldots, V_m \subset \Gamma(\mathbf{P}^n_k, \mathcal{O}(1))$ such that if we take s outside of $\bigcup V_i$, then multiplication by s is injective over $D_+(T_0)$. Similarly for the restriction to $D_+(T_j)$ for $j = 1, \ldots, n$. Since k is infinite, a finite union of proper sub vector spaces is never equal to the whole space, hence we may choose s such that the map is injective. The cokernel of $\mathcal{F}(-1) \to \mathcal{F}$ is annihilated by $\mathrm{Im}(s : \mathcal{O}(-1) \to \mathcal{O})$ which is the ideal sheaf of H by Divisors, Definition 0.3. Hence we obtain $\mathcal{G}$ on H using Cohomology of Schemes, Lemma lemma i star equivalence (Tag 087T). □

Lemma 0.2. *Let k be a field. Let X be a proper scheme over k. Let $\mathcal{L}$ be an ample invertible $\mathcal{O}_X$-module. Let $\mathcal{F}$ be a coherent $\mathcal{O}_X$-module. If $\mathrm{Ass}(\mathcal{F})$ does not contain any closed points, then $\Gamma(X, \mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{L}^{\otimes n}) = 0$ for $n \ll 0$.*

证明. For a coherent $\mathcal{O}_X$-module $\mathcal{F}$ let $\mathcal{P}(\mathcal{F})$ be the property: there exists an $n_0 \in \mathbf{Z}$ such that for $n \leq n_0$ every section s of $\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{L}^{\otimes n}$ has support consisting only of closed points. Since $\mathrm{Ass}(\mathcal{F}) = \mathrm{Ass}(\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{L}^{\otimes n})$ we see that it suffices to prove $\mathcal{P}$ holds for all coherent modules on X. To do this we will prove that conditions (1), (2), and (3) of Cohomology of Schemes, Lemma lemma property higher rank cohomological (Tag 01YM) are satisfied.

To see condition (1) suppose that

$$0 \to \mathcal{F}_1 \to \mathcal{F} \to \mathcal{F}_2 \to 0$$

is a short exact sequence of coherent $\mathcal{O}_X$-modules such that we have $\mathcal{P}$ for $\mathcal{F}_i$, $i = 1, 2$. Let n_1, n_2 be the cutoffs we find. Let $\mathcal{F}_2' \subset \mathcal{F}_2$ be the maximal coherent submodule whose support is a finite set of closed points. Let $\mathcal{I} \subset \mathcal{O}_X$ be the annihilator of $\mathcal{F}_2'$. Since $\mathcal{L}$ is ample, we can find an $e > 0$ such that $\mathcal{I} \otimes_{\mathcal{O}_X} \mathcal{L}^{\otimes e}$ is globally generated. Set $n_0 = \min(n_2, n_1 - e)$. Let $n \leq n_0$ and let t be a global section of $\mathcal{F} \otimes \mathcal{L}^{\otimes n}$. The image of t in $\mathcal{F}_2 \otimes \mathcal{L}^{\otimes n}$ falls into $\mathcal{F}_2' \otimes \mathcal{L}^{\otimes n}$ because $n \leq n_2$. Hence for any $s \in \Gamma(X, \mathcal{I} \otimes_{\mathcal{O}_X} \mathcal{L}^{\otimes e})$ the product $t \otimes s$ lies in $\mathcal{F}_1 \otimes \mathcal{L}^{\otimes n+e}$. Thus $t \otimes s$ has support contained in the finite set of closed points in $\mathrm{Ass}(\mathcal{F}_1)$ because $n + e \leq n_1$. Since by our choice of e we may choose s invertible in any point not in the support of $\mathcal{F}_2'$ we conclude that the support of t is contained in the union of the finite set of closed points in $\mathrm{Ass}(\mathcal{F}_1)$ and the finite set of closed points in $\mathrm{Ass}(\mathcal{F}_2)$. This finishes the proof of condition (1).

Condition (2) is immediate.

For condition (3) we choose $\mathcal{G} = \mathcal{O}_Z$. In this case, if Z is a closed point of X, then there is nothing the show. If $\dim(Z) > 0$, then we will show that $\Gamma(Z, \mathcal{L}^{\otimes n}|_Z) = 0$ for $n < 0$. Namely, let s be a nonzero section of a negative power of $\mathcal{L}|_Z$. Choose a nonzero section t of a positive power of $\mathcal{L}|_Z$ (this is possible as $\mathcal{L}$ is ample, see Properties, Proposition proposition characterize ample (Tag 01Q3)). Then $s^{\deg(t)} \otimes t^{\deg(s)}$ is a nonzero global section of $\mathcal{O}_Z$ (because Z is integral) and hence a unit (Lemma lemma proper geometrically reduced global sections (Tag 0BUG)). This implies that t is a trivializing section of a positive power of $\mathcal{L}$. Thus the function $n \mapsto \dim_k \Gamma(X, \mathcal{L}^{\otimes n})$ is bounded on an infinite set of positive integers which contradicts asymptotic Riemann-Roch (Proposition proposition asymptotic riemann roch (Tag 0BJ8)) since $\dim(Z) > 0$. □

Definition 0.3. Let X be a scheme. Let $\mathcal{L}$ be an invertible sheaf. Let $s \in \Gamma(X, \mathcal{L})$ be a global section. The *zero scheme* of s is the closed subscheme $Z(s) \subset X$ defined by the quasi-coherent sheaf of ideals $\mathcal{I} \subset \mathcal{O}_X$ which is the image of the map $s : \mathcal{L}^{\otimes -1} \to \mathcal{O}_X$.

Primary problem: Tag 0FD8

Lemma 0.4 (Enriques-Severi-Zariski)**.** *Let k be a field. Let X be a proper scheme over k. Let $\mathcal{L}$ be an ample invertible $\mathcal{O}_X$-module. Let $\mathcal{F}$ be a coherent $\mathcal{O}_X$-module. Assume that for $x \in X$ closed we have depth$(\mathcal{F}_x) \geq 2$. Then $H^1(X, \mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{L}^{\otimes m}) = 0$ for $m \ll 0$.*

证明. Choose a closed immersion $i : X \to \mathbf{P}^n_k$ such that $i^*\mathcal{O}(1) \cong \mathcal{L}^{\otimes e}$ for some $e > 0$ (see Morphisms, Lemma lemma finite type over affine ample very ample (Tag 01VT)). Then it suffices to prove the lemma for

$$\mathcal{G} = i_*(\mathcal{F} \oplus \mathcal{F} \otimes \mathcal{L} \oplus \ldots \oplus \mathcal{F} \otimes \mathcal{L}^{\otimes e-1}) \quad \text{and} \quad \mathcal{O}(1)$$

on $\mathbf{P}^n_k$. Namely, we have

$$H^1(\mathbf{P}^n_k, \mathcal{G}(m)) = \bigoplus\nolimits_{j=0,\ldots,e-1} H^1(X, \mathcal{F} \otimes \mathcal{L}^{\otimes j+me})$$

by Cohomology of Schemes, Lemma lemma relative affine cohomology (Tag 089W). Also, if $y \in \mathbf{P}^n_k$ is a closed point then $\text{depth}(\mathcal{G}_y) = \infty$ if $y \notin i(X)$ and $\text{depth}(\mathcal{G}_y) = \text{depth}(\mathcal{F}_x)$ if $y = i(x)$ because in this case $\mathcal{G}_y \cong \mathcal{F}_x^{\oplus e}$ as a module over $\mathcal{O}_{\mathbf{P}^n_k, x}$ and we can use for example Algebra, Lemma lemma depth goes down finite (Tag 0AUK) to get the equality.

Assume $X = \mathbf{P}^n_k$ and $\mathcal{L} = \mathcal{O}(1)$ and k is infinite. Choose $s \in H^0(\mathbf{P}^1_k, \mathcal{O}(1))$ which determines an exact sequence

$$0 \to \mathcal{F}(-1) \xrightarrow{s} \mathcal{F} \to \mathcal{G} \to 0$$

as in Lemma 0.1. Since the map $\mathcal{F}(-1) \to \mathcal{F}$ is affine locally given by multiplying by a nonzerodivisor on $\mathcal{F}$ we see that for $x \in \mathbf{P}^n_k$ closed we have $\text{depth}(\mathcal{G}_x) \geq 1$, see Algebra, Lemma lemma depth drops by one (Tag 090R). Hence by Lemma 0.2 we have $H^0(\mathcal{G}(m)) = 0$ for $m \ll 0$. Looking at the long exact sequence of cohomology after twisting (see Remark remark exact sequence induction cohomology (Tag 0EGK)) we find that the sequence of numbers

$$\dim H^1(\mathbf{P}^n_k, \mathcal{F}(m))$$

stabilizes for $m \leq m_0$ for some integer m_0. Let N be the common dimension of these spaces for $m \leq m_0$. We have to show $N = 0$.

For $d > 0$ and $m \leq m_0$ consider the bilinear map

$$H^0(\mathbf{P}^n_k, \mathcal{O}(d)) \times H^1(\mathbf{P}^n_k, \mathcal{F}(m-d)) \longrightarrow H^1(\mathbf{P}^n_k, \mathcal{F}(m))$$

By linear algebra, there is a codimension $\leq N^2$ subspace $V_m \subset H^0(\mathbf{P}^n_k, \mathcal{O}(d))$ such that multiplication by $s' \in V_m$ annihilates $H^1(\mathbf{P}^n_k, \mathcal{F}(m-d))$. Observe that for $m' < m \leq m_0$ the diagram

$$\begin{array}{ccc} H^0(\mathbf{P}^n_k, \mathcal{O}(d)) \times H^1(\mathbf{P}^n_k, \mathcal{F}(m'-d)) & \longrightarrow & H^1(\mathbf{P}^n_k, \mathcal{F}(m')) \\ \downarrow{\scriptstyle 1 \times s^{m'-m}} & & \downarrow{\scriptstyle s^{m'-m}} \\ H^0(\mathbf{P}^n_k, \mathcal{O}(d)) \times H^1(\mathbf{P}^n_k, \mathcal{F}(m-d)) & \longrightarrow & H^1(\mathbf{P}^n_k, \mathcal{F}(m)) \end{array}$$

commutes with isomorphisms going vertically. Thus $V_m = V$ is independent of $m \leq m_0$. For $x \in \text{Ass}(\mathcal{F})$ set $Z = \overline{\{x\}}$. For d large enough the linear map

$$H^0(\mathbf{P}^n_k, \mathcal{O}(d)) \to H^0(Z, \mathcal{O}(d)|_Z)$$

has rank $> N^2$ because $\dim(Z) \geq 1$ (for example this follows from asymptotic Riemann-Roch and ampleness $\mathcal{O}(1)$; details omitted). Hence we can find $s' \in V$ such that s' does not vanish in any associated point of $\mathcal{F}$ (use that the set of associated points is finite). Then we obtain

$$0 \to \mathcal{F}(-d) \xrightarrow{s'} \mathcal{F} \to \mathcal{G}' \to 0$$

and as before we conclude as before that multiplication by s' on $H^1(\mathbf{P}^n_k, \mathcal{F}(m-d))$ is injective for $m \ll 0$. This contradicts the choice of s' unless $N = 0$ as desired.

We still have to treat the case where k is finite. In this case let K/k be any infinite algebraic field extension. Denote $\mathcal{F}_K$ and $\mathcal{L}_K$ the pullbacks of $\mathcal{F}$ and $\mathcal{L}$ to $X_K = \text{Spec}(K) \times_{\text{Spec}(k)} X$. We have

$$H^1(X_K, \mathcal{F}_K \otimes \mathcal{L}_K^{\otimes m}) = H^1(X, \mathcal{F} \otimes \mathcal{L}^{\otimes m}) \otimes_k K$$

by Cohomology of Schemes, Lemma lemma flat base change cohomology (Tag 02KH). On the other hand, a closed point x_K of X_K maps to a closed point x of X because K/k is an algebraic extension. The ring map $\mathcal{O}_{X,x} \to \mathcal{O}_{X_K,x_K}$ is flat (Lemma lemma change fields flat (Tag 0C4Y)). Hence we have

$$\text{depth}(\mathcal{F}_{x_K}) = \text{depth}(\mathcal{F}_x \otimes_{\mathcal{O}_{X,x}} \mathcal{O}_{X_K,x_K}) \geq \text{depth}(\mathcal{F}_x)$$

by Algebra, Lemma lemma apply grothendieck module (Tag 0338) (in fact equality holds here but we don't need it). Therefore the result over k follows from the result over the infinite field K and the proof is complete. $\square$